

Even Outdegree Edges

Time limit: 1.00 s

Memory limit: 512 MB

Given an undirected graph, your task is to choose a direction for each edge so that in the resulting directed graph each node has an even outdegree. The outdegree of a node is the number of edges coming out of that node.

Input

The first input line has two integers n and m : the number of nodes and edges. The nodes are numbered $1, 2, \dots, n$.

After this, there are m lines describing the edges. Each line has two integers a and b : there is an edge between nodes a and b .

You may assume that the graph is simple, i.e., there is at most one edge between any two nodes and every edge connects two distinct nodes.

Output

Print m lines describing the directions of the edges. Each line has two integers a and b : there is an edge from node a to node b . You can print any valid solution.

If there are no solutions, only print IMPOSSIBLE.

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq m \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$

Example

Input	Output
4 4 1 2 2 3 3 4 1 4	1 2 3 2 3 4 1 4